

CFD Analysis of Aerodynamic Design for Car



Fluid Mechanics II (ME 322)

Instructor: Mr. Faheem Ahmed

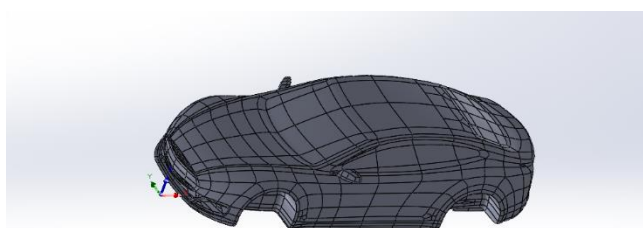
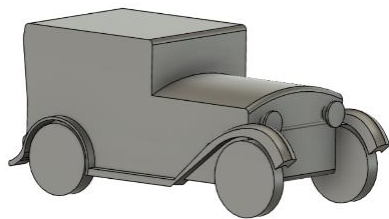
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Abstract

This report presents a comprehensive Computational Fluid Dynamics (CFD) analysis of three distinct cars, aiming to portray the evolution of car design over the years. The analysis primarily focuses on comparing the design characteristics based on the coefficient of drag (C_d) and drag force on the vehicles at various speeds. By conducting the CFD simulations, we were able to investigate the aerodynamic performance of each car and understand the impact of design changes on its overall efficiency. The three cars selected for this study represent different eras in automotive design: Car A, a classic model produced in the 1903s; Car B, a representative modern sedan of the 2010s; and Car C, a modern-day electric vehicle manufactured in the 2020s. Through the CFD analysis, we evaluated the C_d values and drag forces at multiple speeds, ranging from low to high, to capture a comprehensive view of each car's aerodynamic behavior.

Our findings revealed a clear evolution in car design concerning aerodynamics. Car A, with its vintage design, exhibited relatively high C_d values and drag forces across all tested speeds. In contrast, Car B demonstrated improvements in its aerodynamic efficiency compared to Car A, as evidenced by reduced C_d values and drag forces. Finally, Car C, the contemporary vehicle, showcased the most advanced aerodynamic design, significantly reducing both the C_d values and drag forces compared to the previous two cars. The report discusses the underlying factors influencing the aerodynamic performance of each car, including body shape, frontal area, streamlined contours, and various design elements that have evolved. Furthermore, it highlights the role of technological advancements and computational tools in aiding the design process to achieve better aerodynamic efficiency. The results and analysis presented in this report provide valuable insights into the progression of car design over the years and demonstrate the significance of aerodynamics in achieving fuel efficiency and performance. This research contributes to a deeper understanding of how advancements in car design have led to improved aerodynamic characteristics, helping automakers optimize vehicle performance and reduce energy consumption in an increasingly sustainable world.



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1. Introduction

The automotive industry has been rapidly evolving over the past century, constantly seeking ways to improve vehicle performance, safety, and fuel efficiency. One pivotal element in achieving these goals is the aerodynamic design of the car. It significantly influences the car's fuel consumption, stability at high speeds, engine cooling, passenger comfort, and even the overall aesthetic appeal. Traditional approaches to studying and optimizing aerodynamics involve wind tunnel testing, which can be both time and cost-intensive. However, in the age of digital transformation, we have more efficient tools at our disposal, one of the most prominent being Computational Fluid Dynamics (CFD).

This report presents a comprehensive CFD analysis of car designs, a sophisticated, computer-based method for predicting fluid flow, heat transfer, and other related phenomena. CFD utilizes numerical methods and algorithms to solve and analyze fluid flow issues. Computers are used to perform the calculations required to simulate the interaction of fluids and gases with the complex surfaces used in car design.

By applying CFD, we can virtually prototype and test the design of a car, making necessary adjustments to enhance its aerodynamic properties. This technique allows us to optimize design performance, reduce physical prototyping costs, and shorten design cycles. It not only helps in reducing the drag but also helps in improving the downforce, cooling effects, and reducing lift, thus impacting the overall performance of the vehicle.

In this report, we will delve into the specifics of using CFD to simulate and analyze the aerodynamics of a car design. We will discuss the methodology employed, the benefits of this approach, the challenges encountered, and the insights gained from the simulation results. Our focus will be on how over the years, the design of road vehicles has revolutionized to incorporate more aerodynamic design and how these design changes have affected the drag force applied on the vehicle as they travel at different velocities.

2. Literature Review

The aerodynamic design is a critical factor in maximizing the performance and efficiency of automobiles. By implementing effective aerodynamic principles, designers can minimize drag, improve fuel efficiency, and enhance speed. The shape of the car plays a pivotal role in its aerodynamic performance. A streamlined shape with smooth contours helps reduce drag, leading to better fuel efficiency. Designers must carefully consider the car's overall shape, including its body's angles and curves, to optimize aerodynamics. The underbody design of the car is another critical aspect that influences aerodynamics. A smooth underbody with minimal protrusions is essential to minimize drag and increase fuel efficiency. Designers need to focus on creating a streamlined underbody by ensuring smooth surfaces and minimizing obstructions that disrupt airflow. [1]

The frontal area of the car significantly impacts aerodynamics. A smaller frontal area reduces air resistance, directly contributing to improved fuel efficiency. Designers should aim to minimize the height and width of the car's front end to decrease the frontal area and enhance aerodynamic performance. Good aerodynamic design requires meticulous attention to detail and a thorough understanding of airflow. By optimizing the car's shape, frontal area, underbody design, and other key features, designers can create efficient, fast, and stable cars.

The design of cars has evolved significantly over the years concerning aerodynamics. While the earliest cars paid little attention to aerodynamic principles, engineers and designers gradually began to

understand and utilize these principles to improve vehicle performance, fuel efficiency, and stability. Here's a brief overview of the evolution:

- Early Cars (The late 1800s - Early 1900s): The earliest cars were not designed with aerodynamics in mind. They were typically boxy and bulky, leading to high levels of drag.
- Streamlined Cars (1930s - 1940s): As engineers began to understand the principles of aerodynamics, car designs started to become more streamlined. The Chrysler Airflow, introduced in 1934, was one of the first cars designed with aerodynamics as a key consideration.
- Post-War Era (1950s - 1960s): The post-WWII era saw a boom in car manufacturing, with designs becoming more distinctive and diverse. Still, aerodynamics was not a primary concern for many car manufacturers.
- Fuel Crisis and the Rise of Aerodynamics (1970s): The oil crisis of the 1970s led manufacturers to focus on fuel efficiency. Cars became more streamlined to reduce drag and increase fuel economy. This period saw the introduction of cars like the Audi 100, which had a low drag coefficient for its time.
- Computer-aided Design (1980s - 1990s): With the advent of computer-aided design (CAD) and wind tunnel testing, engineers were able to make more precise calculations about airflow and design cars to be more aerodynamically efficient. Cars like the Honda CRX and GM EV1 were notable for their low drag coefficients.
- Modern Era (2000s - Present): Today, nearly every aspect of a car's design is optimized for aerodynamics, from the shape of the body to the placement of the mirrors. Hybrid and electric vehicles, in particular, focus heavily on aerodynamics to maximize range. Cars like the Tesla Model S and the Mercedes-Benz CLA are known for their extremely low drag coefficients.

In the future, we can expect car designs to continue evolving with a focus on improving aerodynamics even further. Concepts such as active aerodynamics, where elements of the car adjust dynamically to changing conditions, are likely to become more prevalent. [2]

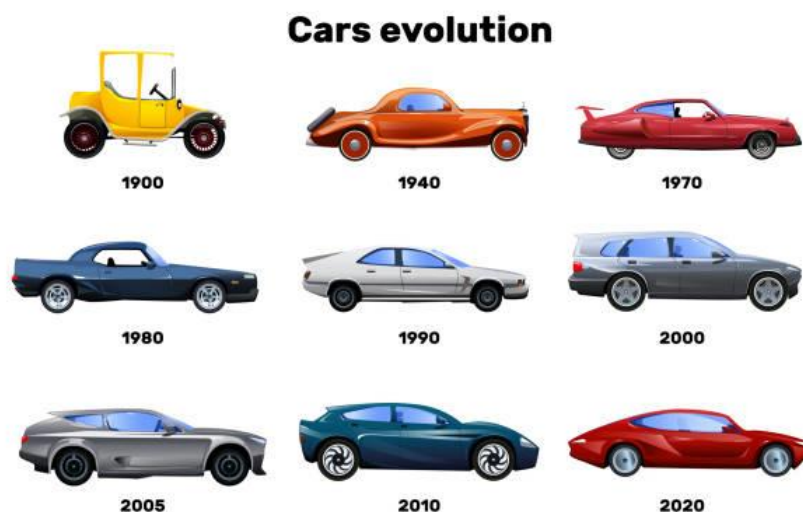


Figure 1 Evolution of Cars

CFD (Computational Fluid Dynamics) analysis is a valuable tool employed in engineering and design to simulate and study fluid behavior, such as air or water, in different systems. Its applications span across

industries, including aerospace, automotive, and manufacturing, offering insights and solutions for various engineering challenges.

This literature review focuses on the application of Computational Fluid Dynamics (CFD) analysis in the context of aerodynamic design for automobiles. The study aims to explore the recent advancements, methodologies, and challenges associated with CFD analysis for improving the aerodynamic performance of vehicles. Through an extensive review of scholarly articles, this literature review provides an overview of key concepts, approaches, and findings, ultimately highlighting the importance of CFD in enhancing vehicle efficiency, reducing fuel consumption, and minimizing environmental impact.[3]

When a vehicle is in motion at a particular velocity, the fluid experiences viscous effects primarily within a thin layer called the boundary layer. Beyond this boundary layer, the flow is considered inviscid, as viscous effects are negligible. The pressure on the edge of the boundary layer is exerted by the fluid flow. At the rear end of the vehicle, the fluid flow can separate.

The behavior of the fluid within the boundary layer is governed by its viscosity. For Reynolds numbers below the 10^4 , a distinct boundary layer does not exist. The Reynolds number depends on the characteristic length of the vehicle, the kinematic viscosity of the fluid, and the velocity of the vehicle. In other words, the flow of fluid around the vehicle is influenced by the vehicle's shape and the Reynolds number.

Another important phenomenon that affects the flow of fluid past the vehicle body and the vehicle's performance is known as the "wake." When the airflow over the vehicle separates at the rear end, it creates a turbulent region of low pressure behind the vehicle, referred to as the wake. This wake contributes to pressure drag, which in turn reduces the overall performance of the vehicle.

In Computational Fluid Dynamics (CFD), the solver plays a crucial role in solving and analyzing the model based on the given boundary conditions. It is important to carefully define the solver steps, which can vary depending on the type of analysis being performed. The solver setup is a time-consuming aspect of CFD analysis, and it involves considerations such as the size of the generated mesh. [4]

Engineers and designers rely on CFD (Computational Fluid Dynamics) analysis to optimize system designs by predicting their performance under various conditions, enabling them to assess and enhance the aerodynamic performance of car designs to improve efficiency, decrease drag, and boost speed. This is inclusive of the aerodynamics of automobiles, optimizing of heat exchangers for improved efficiency, and simulation of airflow around aircraft wings, consequently tuning parameters of the car's body shape, windshield angle, and side mirror design to minimize drag and maximize fuel efficiency. By leveraging CFD analysis, engineers can make informed design decisions leading to the development of cars.

The mesh size directly impacts the solver's computational time. A finer mesh requires more time to solve the problem because the equations are solved for each element. With smaller element sizes, the number of elements increases for a given model. Therefore, selecting an appropriate mesh size is crucial for balancing accuracy and computational efficiency. [5]

The solver setup typically includes the following steps:

- **Type of Analysis:** For aerodynamic analysis, a 3D model analysis is carried out using a pressure-based solver with relative velocity formulation. Conclusively, a steady-state analysis is done.
- **Models:** The numerical model used for the analysis is defined here. For aerodynamic analysis, a viscous-based numerical model is employed.
- **Materials:** The material for the model is defined in this step. In the case of analyzing air as the surrounding fluid, the properties of air can be retrieved from the ANSYS Library.
- **Boundary Conditions:** Defining the initial boundary conditions is a crucial part of the ANSYS Solver. These variables are considered initial boundary values, as they are necessary to initialize the solution.
- **Monitors:** Monitoring the required values of drag coefficient (C_d) and lift coefficient (C_l) is important. These values can be used to assess the accuracy and convergence of the results.

After performing these initial setups, the calculation is run for a required number of iterations. Increasing the number of iterations brings the obtained results closer to the actual values. The convergence of the C_d and C_l plots can be used to determine the accuracy of the results.

Applications of CFD Analysis

- Here are some applications of CFD analysis in the aerodynamic design of automobiles:
- CFD Analysis can assist in improving a vehicle's design to lower drag and increase fuel efficiency.
- CFD analysis can be used to optimize a vehicle's design for stability at high speeds by examining the airflow over the vehicle.
- CFD analysis can help in lowering wind noise, by examining the airflow surrounding the vehicle and locating regions of significant turbulence.
- CFD Analysis is used significantly in the racing industry to enhance the aerodynamics of racing cars.

For the CFD analysis of the vehicle models the parameters we will be judging are the coefficient of drag (C_d) and the drag force both of these parameters are dependent on different physical properties of the vehicle as well as the properties of the fluid.

- **Drag force** on a vehicle is influenced by several factors. Here are some of the primary ones:
- **Shape and Size of the Vehicle (Vehicle Aerodynamics):** The shape of the vehicle is the most significant factor affecting drag. Aerodynamically efficient designs have a streamlined shape that reduces air resistance.
- **Vehicle Speed:** The drag force experienced by a vehicle increases with the square of its speed. Thus, a vehicle moving at high speeds will experience more drag than the same vehicle moving at lower speeds.
- **Air Density:** The drag force is directly proportional to the air density. Therefore, a vehicle will experience more drag at lower altitudes (where the air is denser) than at higher altitudes.
- **Frontal Area:** The larger the frontal area (the area of the vehicle facing the wind), the greater the drag. This is because a larger frontal area will collide with more air molecules, leading to increased resistance.
- **Surface Roughness:** Smooth surfaces reduce friction and hence the drag force. On the other hand, rough surfaces increase the friction between the vehicle's surface and the air, leading to higher drag.

- **Flow Conditions:** Turbulent air flow creates more drag than laminar flow. Design features that promote a smooth, laminar flow of air over the vehicle can help to reduce drag.
- **Tire Pressure:** Lower tire pressures can increase the contact area between the tire and the road, leading to increased rolling resistance and, indirectly, increased drag.
- **Wind Speed and Direction:** The speed and direction of the wind can significantly impact the drag force on a vehicle. Headwinds increase the effective speed of the vehicle relative to the air, thus increasing drag, while tailwinds decrease it. Crosswinds can also increase drag by creating pressure differentials around the vehicle.

Each of these factors contributes to the overall drag experienced by a vehicle. Reducing drag can lead to improvements in fuel efficiency and vehicle performance. [6]

- **The coefficient of drag (C_d)** of a vehicle is influenced by several factors. Here are some of the key elements that determine the drag coefficient:
- **Shape and Size:** The aerodynamics of a vehicle, which is directly related to its shape and size, is one of the biggest factors affecting the coefficient of drag. A smooth, streamlined design will have a lower C_d than a boxy design.
- **Surface Roughness:** The smoother the surface, the lower the drag. This is why vehicles are painted and polished, to reduce the surface roughness and thus reduce the drag.
- **Speed:** The faster a vehicle moves, the more air it displaces and the greater the drag. However, the drag coefficient itself is independent of speed; it is a property of the shape and surface of the object, not how fast the object is moving.
- **Air Density:** The denser the air, the higher the drag. Air density can be affected by altitude (air is denser at sea level than at high altitudes) and by temperature (cold air is denser than warm air).
- **Frontal Area:** The larger the frontal area that is exposed to the oncoming air, the higher the drag. This is why designers try to minimize the frontal area of high-performance cars.
- **Flow Separation:** The point at which the airflow separates from the body of the vehicle also affects the drag. Designers aim to move this separation point toward the rear of the vehicle to reduce drag.
- **Reynolds Number:** This is a measure of the ratio of inertial forces to viscous forces and describes the degree of turbulence or laminar flow. It can affect the drag coefficient, particularly for vehicles moving at high speeds or in very viscous fluids.
- **Vehicle Add-ons:** Any additions to the vehicle like spoilers, mirrors, roof racks, etc., can change the aerodynamics and thus affect the C_d .

Designing vehicles with a low coefficient of drag can greatly improve their fuel efficiency, especially at higher speeds where air resistance becomes a significant factor.[7]

We can conclude from the literature review that the value of the coefficient of drag would range between 0.195-0.45 depending on the velocity of the fluid passing over the model. Apart from this, the value for drag force would vary depending on the velocity as well as the shape and geometry of the vehicle. Streamlined vehicles with smaller surface areas would experience reduced drag as compared to bulky models. [8]

3. Numerical Model

The computational fluid dynamics study was performed on the Ansys Fluent module. The choice between ANSYS Fluent and ANSYS CFX for CFD (Computational Fluid Dynamics) analysis depended on several factors, including the specific application, the characteristics of the flow, and the user's familiarity with the software. The major consideration that we considered was the Physics and Solver Capabilities. ANSYS Fluent and ANSYS CFX have different solver algorithms and are suitable for different types of flow simulations. Fluent is known for its versatility and is well-suited for a wide range of applications, including general-purpose CFD analysis, multiphase flows, combustion, and aerodynamics. CFX, on the other hand, is particularly adept at handling rotating machinery, such as pumps and turbines, as well as multiphase flows and conjugate heat transfer problems. Therefore since our study involved the simulation of the aerodynamic properties of vehicles at different speeds we chose ANSYS Fluent to perform the study.



3.1. Setup

Viscous standard k- ϵ Model

The simplest “complete models” of turbulence are two-equation models in which the solution of two separate transport equations allows the turbulent velocity and length scales to be independently determined. The standard k- ϵ model in FLUENT falls within this class of turbulence models and has become the workhorse of practical engineering flow calculations in the time since it was proposed by Launder and Spalding. Robustness, economy, and reasonable accuracy for a wide range of turbulent flows explain its popularity in industrial flow and heat transfer simulations. It is a semi-empirical model, and the derivation of the model equations relies on phenomenological considerations and empiricism. As the strengths and weaknesses of the standard k- ϵ model have become known, improvements have been made to improve its performance. Two of these variants are available in FLUENT: the RNG k- ϵ model and the realizable k- ϵ model. The standard k- ϵ model is a semi-empirical model based on model transport equations for the turbulence kinetic energy (k) and its dissipation rate (ϵ). The model transport equation for k is derived from the exact equation. In contrast, the model transport equation for ϵ was obtained using physical reasoning and bears little resemblance to its mathematical counterpart. In the k- ϵ model derivation, the assumption is that the flow is fully turbulent, and the effects of molecular viscosity are negligible. The standard k- ϵ model is, therefore, valid only for fully turbulent flows.

Transport Equations for the Standard k- ϵ Model

The turbulence kinetic energy, k , and its rate of dissipation, ϵ , are obtained from the following transport equations:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k + G_b - \rho \epsilon - Y_M + S_k$$

$$\frac{\partial}{\partial t}(\rho \epsilon) + \frac{\partial}{\partial x_i}(\rho \epsilon u_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right] + C_{1\epsilon} \frac{\epsilon}{k} (G_k + C_{3\epsilon} G_b) - C_{2\epsilon} \rho \frac{\epsilon^2}{k} + S_\epsilon$$

In these equations, G_k represents the generation of turbulence kinetic energy due to the mean velocity gradients. Modeling Turbulent Production in the k- ϵ Models. G_b is the generation of turbulence kinetic energy due to buoyancy, the effects of Buoyancy on Turbulence in the k- ϵ Models. Y_M represents the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate, calculated as described in Section 12.4.6: Effects of Compressibility on Turbulence in the k- ϵ Models. C_1 , C_2 , and C_3 are constants. σ_k and σ_ϵ are the turbulent Prandtl numbers for k and ϵ , respectively. S_k and S_ϵ are user-defined source terms. [9]

Spatial discretization

One of the key steps in setting up a Fluent simulation is spatial discretization, which is the process of breaking down the physical domain into a finite number of discrete elements or cells. This process helps in solving the partial differential equations (PDEs) that describe fluid flow. There are several discretization schemes available in Fluent for different types of terms in the governing equations:

- First-Order Upwind: This scheme is based on the direction of the flow. It's a simple and stable method, but it may introduce numerical diffusion, reducing accuracy.
- Second-Order Upwind: This scheme provides a higher level of accuracy than the first-order upwind scheme, but it's less stable.
- QUICK (Quadratic Upstream Interpolation for Convective Kinematics): This is a higher-order scheme providing even more accuracy, but it requires a fine mesh and it's computationally expensive.
- Central Differencing: This scheme is second-order accurate and is used for diffusion terms. It's accurate but can be unstable for convection-dominated flows.
- Power Law: This scheme is used for convection terms and provides a balance between accuracy and stability.

The discretization scheme choice depends on the problem's nature, the mesh's quality, and the required balance between accuracy and computational cost. Since the model for which we obtained the results was relatively simple. For comparison, we needed accurate results, therefore, we went with second-order upwind equations to solve our simulation.[10]

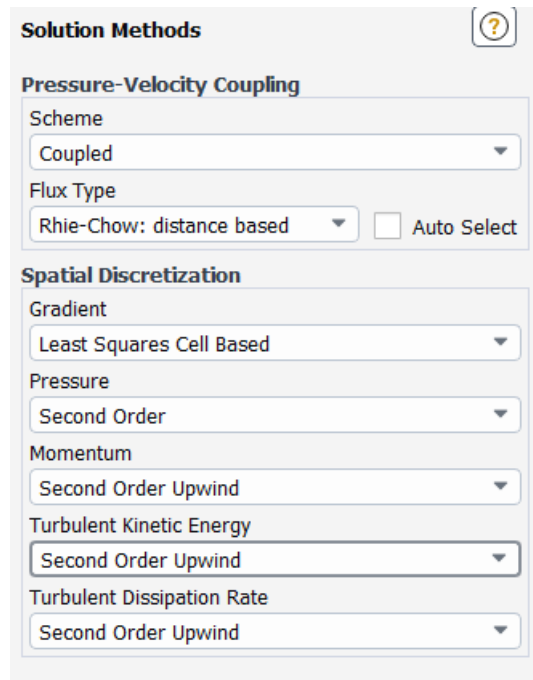


Figure 2 Solution Method

Reference frame and input velocities:

In ANSYS Fluent, reference frames are used to define the motion of your simulation domain. There are two types of reference frames: absolute and relative. The absolute reference frame is fixed and does not move. This is the default frame of reference in Fluent. When you specify an inlet velocity, by default, it's in the absolute reference frame. This means the velocity you set is the actual speed and direction of the fluid entering your domain relative to a stationary observer.

For better understanding and clear visualization while studying the results of the simulation we selected the absolute frame of reference.

The simulation was performed on 3 different air velocities. These velocities portrayed different driving conditions.

- 60 km/hr: daily drive through the city
- 100 km/hr: driving on the highway
- 150 km/hr: driving on the race track.

The turbulent viscosity ratio, also known as the turbulence viscosity ratio or turbulence eddy viscosity ratio, is a dimensionless quantity used in turbulence modeling, especially in the field of computational fluid dynamics (CFD). It's the ratio of turbulent viscosity (μ_t) to the molecular (or laminar) viscosity (μ). In practical terms, the turbulent viscosity ratio is used to adjust the turbulence models in CFD simulations. In ANSYS Fluent, for example, this ratio is often limited to prevent it from becoming too high, which could lead to numerical stability issues. The recommended values for the turbulent viscosity ratio limit in Fluent can vary depending on the problem but commonly suggested values range from 1 to 10. Hence for our simulation, we used a turbulent viscosity ratio of 10.

Velocity Inlet

Zone Name
inlet

Momentum Thermal Radiation Species DPM Multiphase Potential Structure UDS

Velocity Specification Method: Magnitude, Normal to Boundary

Reference Frame: Absolute

Velocity Magnitude [km/h]: 100

Supersonic/Initial Gauge Pressure [Pa]: 0

Turbulence

Specification Method: Intensity and Viscosity Ratio

Turbulent Intensity [%]: 5

Turbulent Viscosity Ratio: 10

Apply Close Help

Figure 3 Inlet Velocity Setup

3.2. Meshing:

3.2.1. Grid Independence:

In computational fluid dynamics (CFD) simulations using ANSYS or any other program, grid independence testing is an essential step. By using a mesh to discretize the computational domain, it seeks to ensure that the simulation results are independent of the mesh's size or spacing.

The following are the steps for running a grid independence test for CFD on ANSYS:

- **Choose a mesh:** Choose a mesh size and spacing that you think will provide a decent solution as a starting point.
- **Execute the simulation:** Run the CFD simulation using the chosen mesh parameters (spacing and size).
- **Increase the number of mesh elements** by fine-tuning the mesh in certain domain regions. Maintain a stable domain size and shape overall.
- **Simulation repetition:** Rerun the simulation with the improved mesh.

Examine the outcomes to see if the answer is affected by the mesh size and spacing. Compare the outcomes of the two simulations. Repeat the approach by further refining the mesh until the solution is grid-independent if the outcomes are noticeably different.

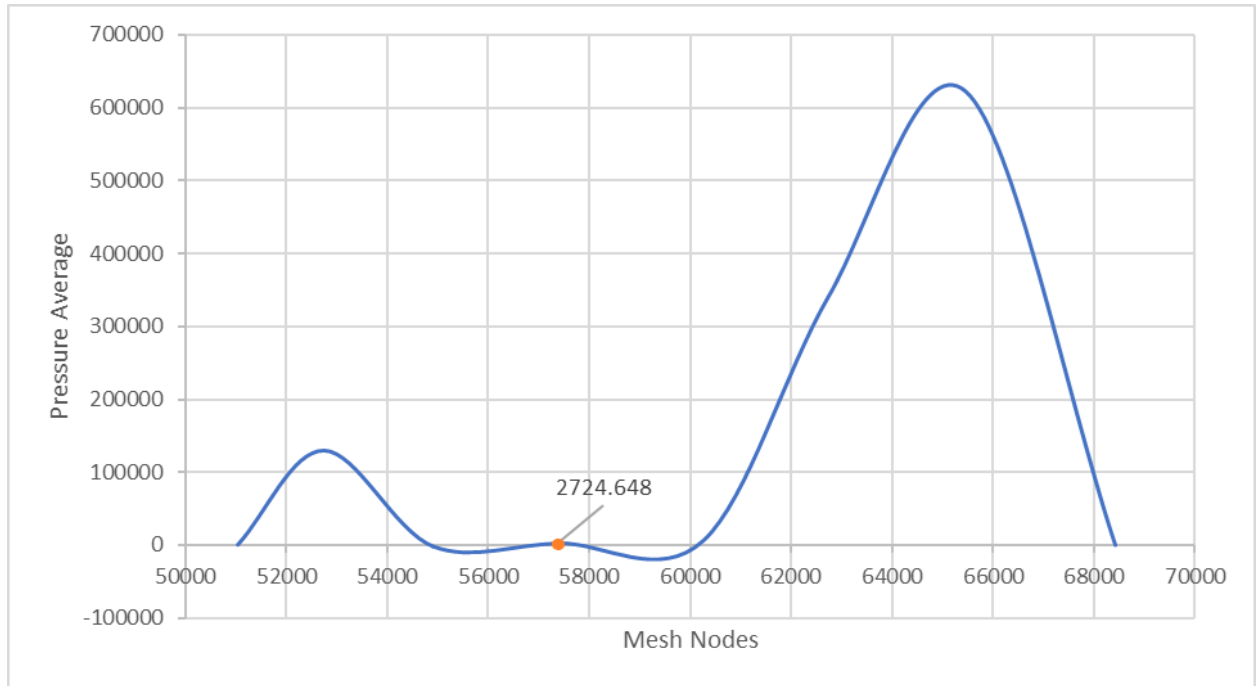


Figure 4 Grid Independence plot

Here we can see from the graph that the only stable region lies between 56000-60000 mesh nodes. Therefore, we can select a point from this range to use as our optimum number of sizing numbers corresponding to the selected mesh nodes. The point selected is at **57396 mesh nodes** i.e., **40 number of divisions**.

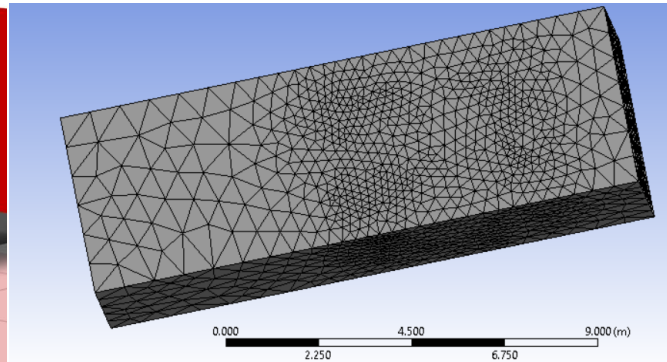
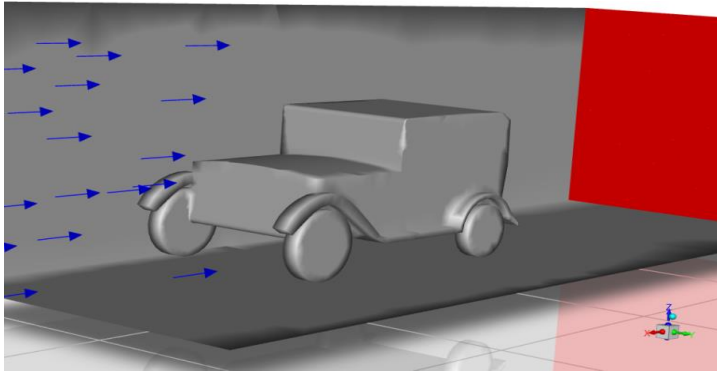
Name	P3 - Edge Sizing Number of Divisions	P1 - Mesh Nodes	P2 - Mesh Elements	P4 - pressure-average [Pa]
DP 0	10	51031	280334	646.72555
DP 1	20	52741	288009	129926.43
DP 2	30	54902	298417	-1655.0037
DP 3	40	57396	310437	2724.648
DP 4	50	60252	324291	6266.1061
DP 5	60	62733	336162	341182.88
DP 6	70	65448	349698	622505.96
DP 7	80	68415	364105	79.333335

Figure 5 Values calculated for Grid Independence

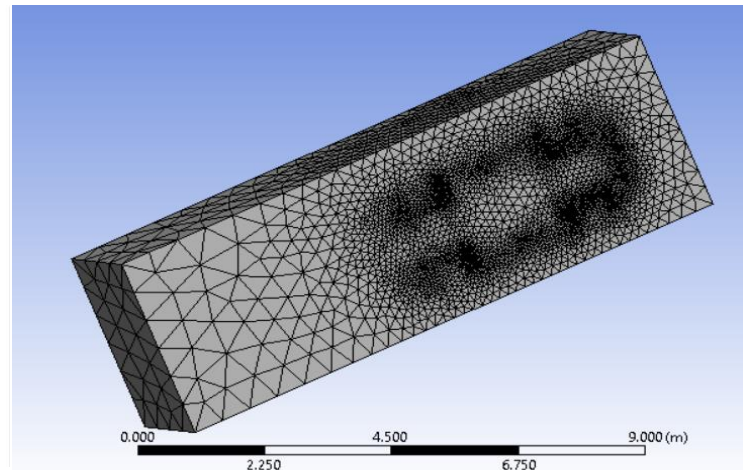
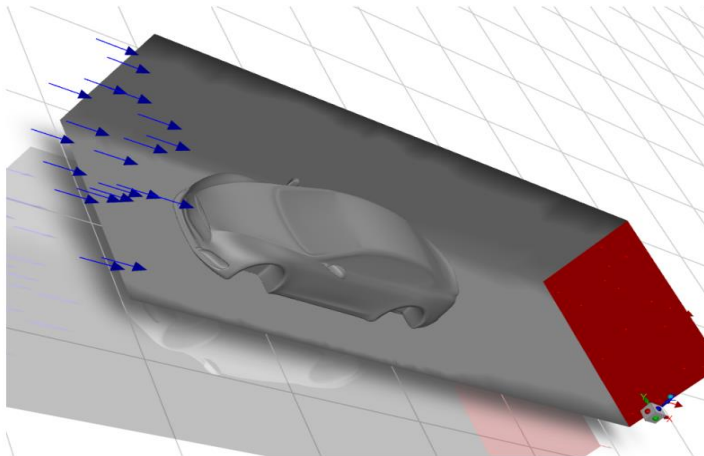
Note: Due to computational power limitations, the grid dependence test was only run on the Box car, and the other models were taken on the assumption to be the same as the box car model.

3.2.2. Final Mesh:

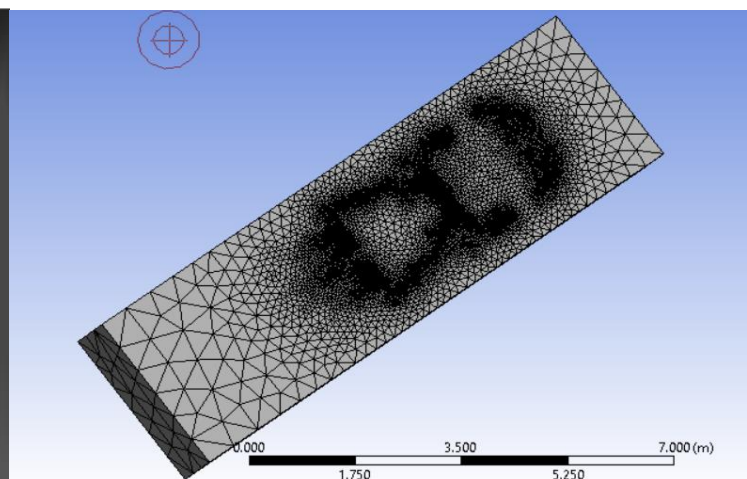
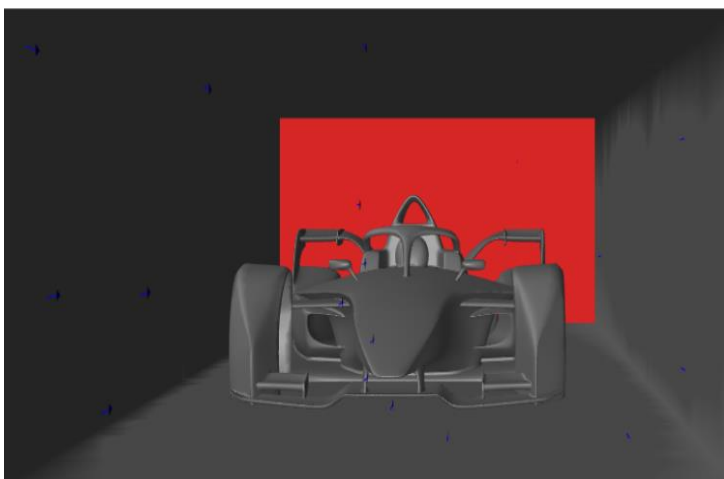
Mesh Divisions: 40



Box Car Nodes = 57396



Tesla Nodes: 378247



Formula Car Nodes: 10399078

4. Computational Solution

To achieve valid and comparable results certain parameters were kept constant for each of the three car models.

- The model of the vehicles were scaled so that their dimensions were comparable to each other. In our study, we made sure that the length of each of the three car models was set to 5 meters. Through this process, we ensured that the size of the vehicle did not affect the results and that only the shape and aerodynamics of the vehicle had any effect on the calculated results.
- All car models were taken to be solid.
- The inlet area for all nine studies was kept constant

CFD simulation was carried out for all three vehicle models at 60, 100, and 150 kilometers per hour of air velocity. Through this process, we expected to get a clear picture of the behavior of different designs at different road speeds and determine the characteristics of each design. For the computation, due to limited computation power available we set different benchmarks for the number of iterations we use to reach our result. The specific details for each model are explained below:

- Box type vehicle: 100 iterations were performed to achieve our results. The processing time for these iterations came out to be 7 minutes.
- Tesla model: 40 iterations were performed to achieve our results. The processing time for these iterations was 2 hours 20 minutes.
- Electric formula vehicle: 14 iterations were performed to achieve our results. The processing time for these iterations was 3 hours 45 minutes.

We generated the solution plots for the given study for the different models. The images of the results are calculated for an air velocity of 100 km/hr. Results for other air velocities were similar, and only the intensity increased; the behavior of the plots remained constant.

4.1. Simulation results for box-type vehicle

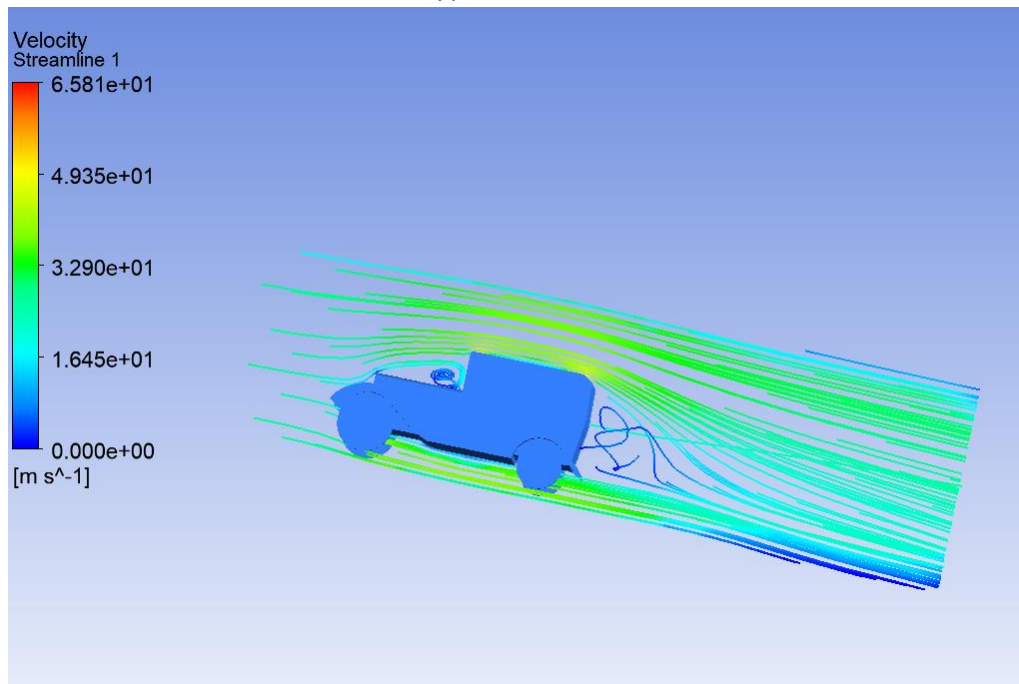


Figure 6 Velocity Stream lines on a midplane

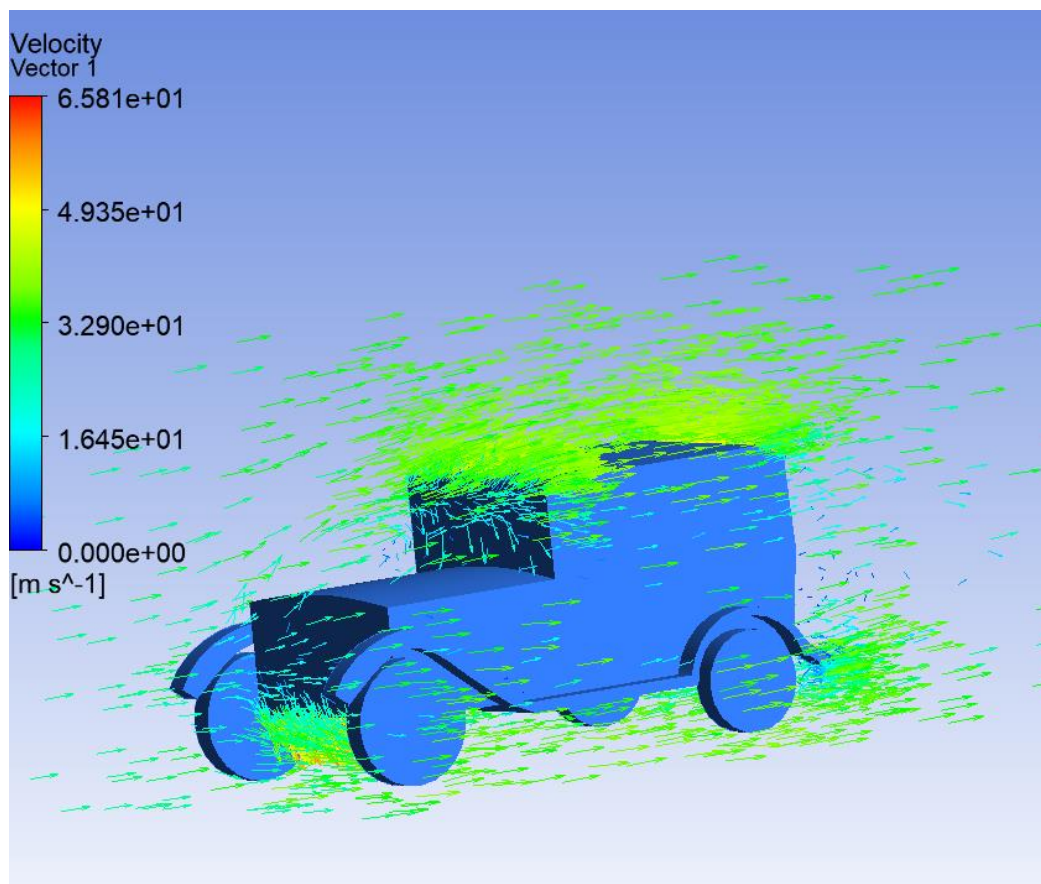


Figure 7 Velocity vectors

4.2. Simulation results for the Tesla model

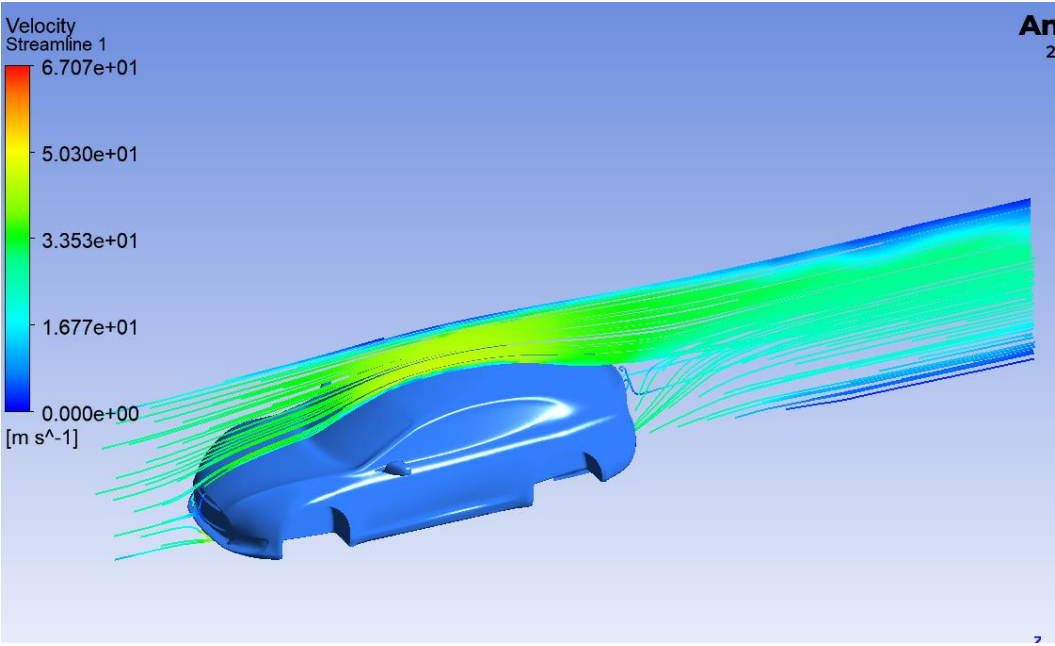


Figure 10 Airflow over the vehicle (midplane)

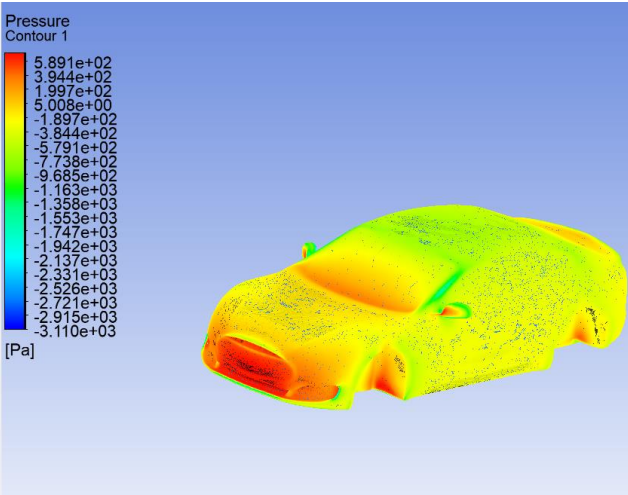


Figure 11 Pressure distribution (car body surface)

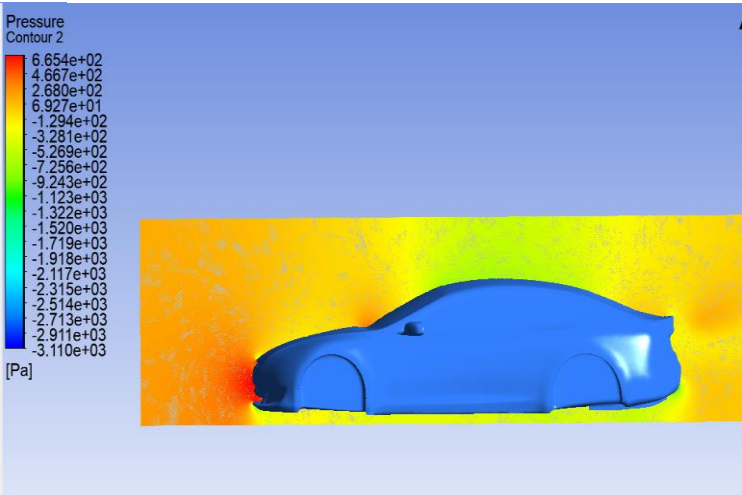


Figure 9 Pressure distribution (midplane)

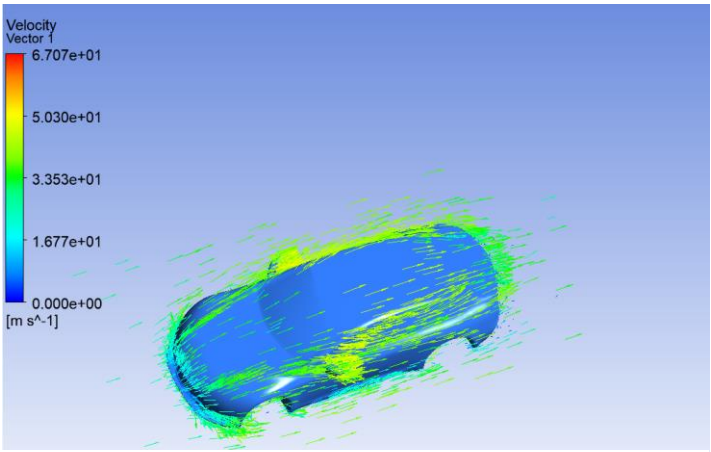


Figure 8 Velocity Distribution (car body surface)

4.3. Simulation results for electric formula vehicle

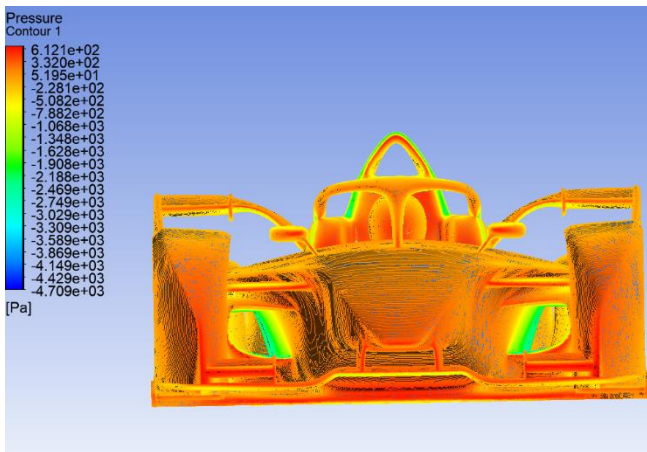


Figure 13 Front Pressure Difference (Car body Surface)

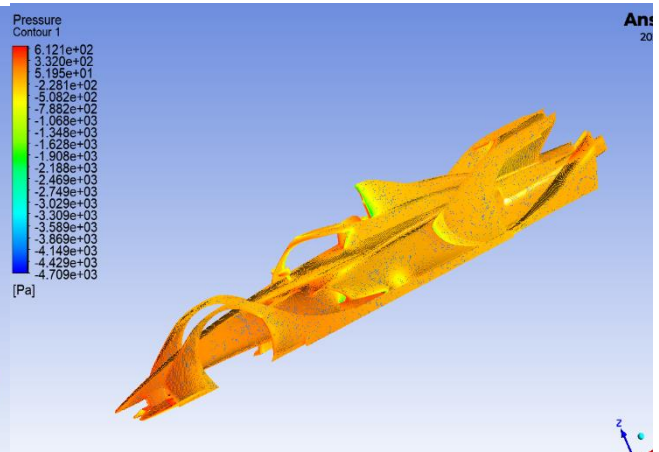


Figure 14 Pressure distribution (Car body Surface)

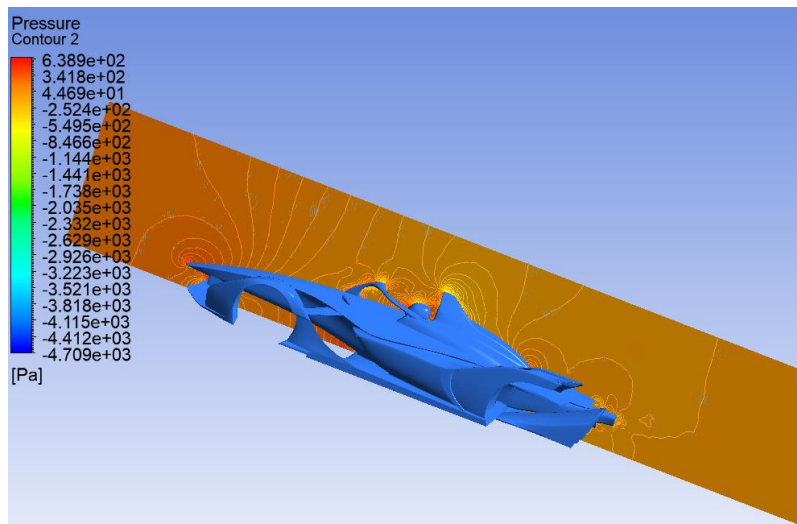


Figure 12 Pressure Distribution Mid Plane

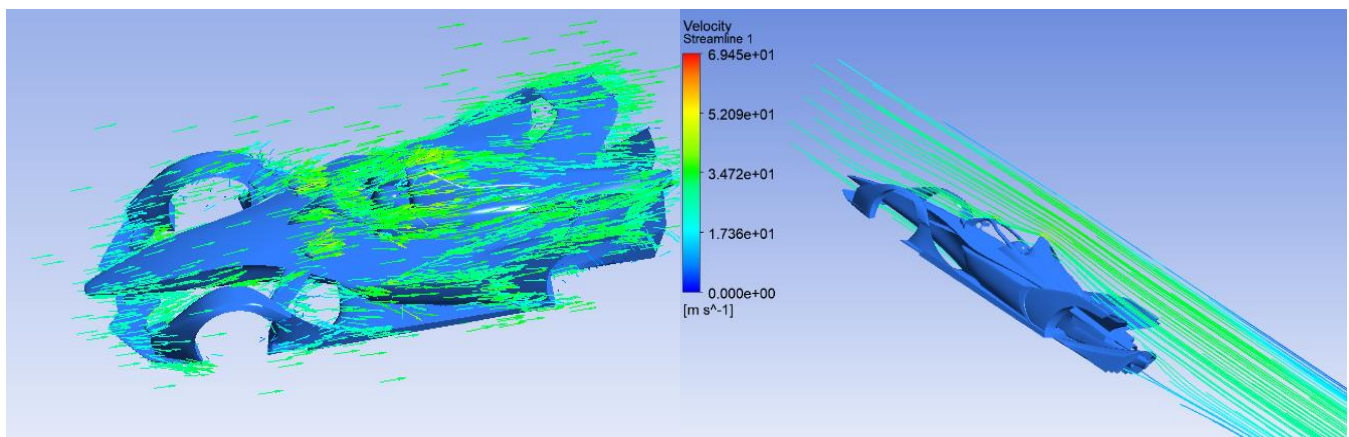


Figure 15 Air flow over Vehicle (Midplane)

Figure 16 Velocity Distribution (car body surface)

5. Analysis of results

The values of drag force and coefficient of drag were computed and plots were made to compare these properties at different velocities for different vehicles.

5.1. Drag Force

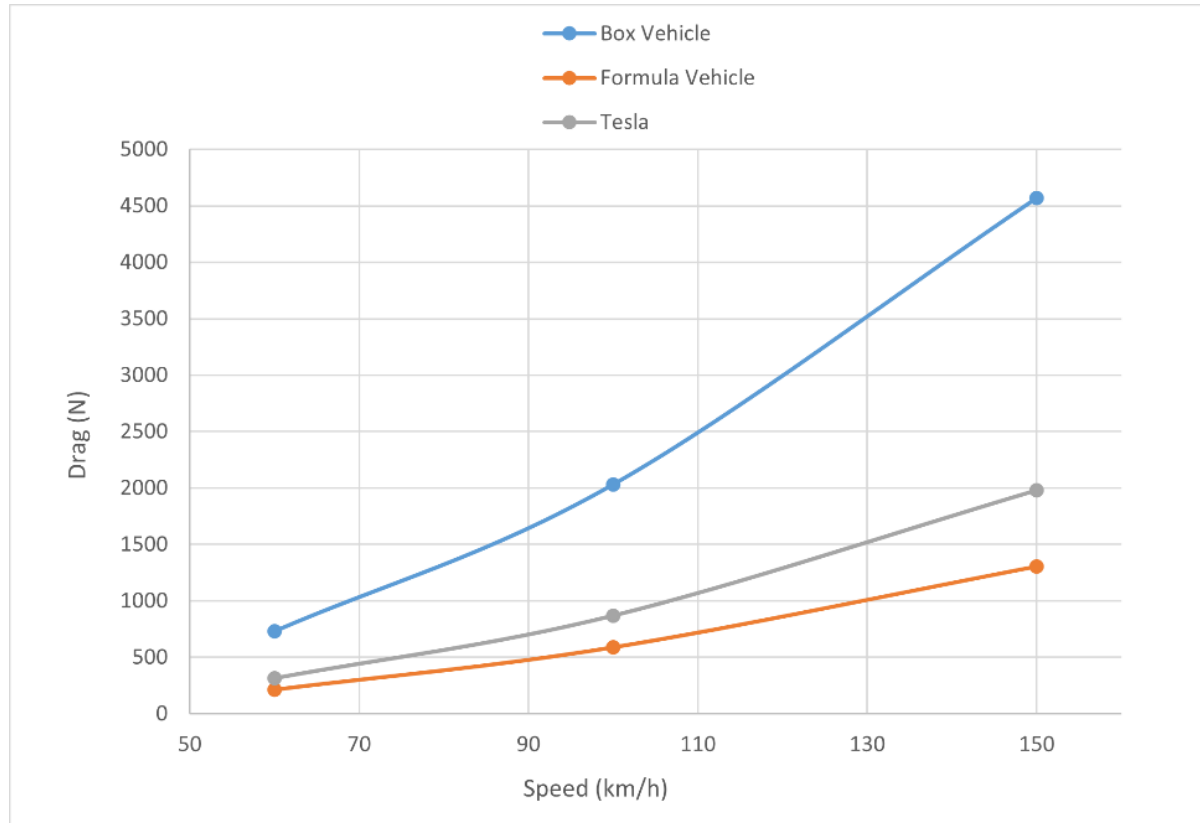


Figure 17 Speed vs Drag Force

As shown above, the plot between drag force and air velocity shows that as the velocity of air increases the drag force on the vehicle increases proportionally. This trend can be justified by as the speed of an object increases, more fluid molecules collide with the object's surface, leading to an increase in pressure and shear stress on the object. This results in a higher drag force acting against the object's motion. From the equation of drag force, we can see that the drag force is proportional to the square of the object's speed (V^2). This means that as the speed increases, the drag force increases quadratically, leading to a significant rise in the force opposing the object's motion.

The difference in drag force of different vehicles is caused by the air pressure differences in front of and behind the vehicle. A blunt object, like a box-type vehicle, experiences a large pressure difference because the air piles up in front of it and creates a vacuum behind it. A streamlined object, like a sport's electric car, reduces this pressure difference by allowing the air to flow more smoothly around it, reducing the pressure drag.

5.2. Coefficient of drag

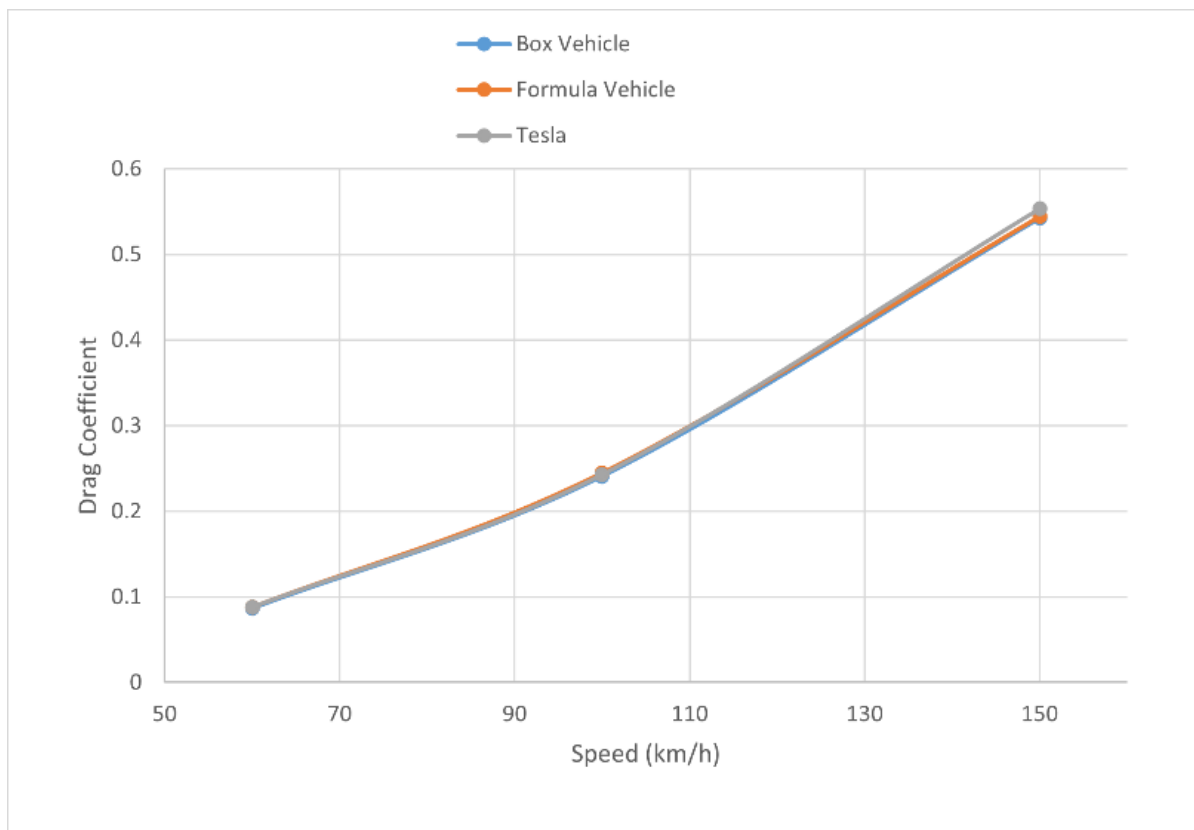


Figure 18 Drag Coefficient vs Speed.

In general, the coefficient of drag is primarily determined by the shape and surface characteristics of the object, not its velocity. However, at different flow regimes characterized by the Reynolds number (which involves velocity), the coefficient of drag can indeed vary. Since the relative shape of each vehicle model is the same, the drag coefficient is approximately the same. The increasing trend of the drag coefficient is due to the increase in the Reynolds number due to the increase in the velocity of air flowing over the vehicle model.

5.3. Data Observed

		Box Vehicle		Formula Vehicle Chassis		Tesla Chassis	
S No.	Speed, U (km/h)	Drag Coefficient, Cd	Drag, D (N)	Drag Coefficient, Cd	Drag, D (N)	Drag Coefficient, Cd	Drag, D (N)
1	60	0.0865	729	0.0881	211.47	0.08771	314.0737
2	100	0.241	2030.49	0.245	588.16	0.2429	869.9
3	150	0.5422	4567.96	0.5443	1306.8	0.553655	1982.33

6. Conclusion

The comprehensive study carried out in this project has provided valuable insights into the performance characteristics of three distinctive vehicles, spanning different eras and design philosophies, through the application of Computational Fluid Dynamics (CFD) analysis. Our investigation focused on the aerodynamic properties of a 1940s box-like model, a Tesla Model S, and an electric Formula race car. The CFD analysis allowed us to examine the impact of vehicular design on aerodynamics, primarily through the lens of drag coefficients (C_d) and overall drag. Our findings underlined that the 1940s box-like model showcased significantly high drag and C_d values while representing its era. This is primarily due to its box-like structure, which is not conducive to airflow and results in a turbulent wake. The Tesla Model S, a contemporary model, fared considerably better with its streamlined design offering reduced drag and a lower C_d value, exemplifying the advancements in automotive design over the years. However, the standout performer was the electric Formula race car. Its specialized design, optimized for speed and efficiency, yielded the lowest drag and C_d values among the three vehicles. The results were in line with what we had studied in our literature review. The trends of drag force followed the outline provided by the literature. Similarly, the value of the coefficient of drag was also like what we expected from our literature review, following a trend according to which the coefficient of drag increases with an increase in velocity.

The meshing process in the CFD simulation was a critical factor in the accuracy of our findings. The complexity of the vehicle shapes, particularly the Formula race car with its intricate aerodynamic elements, necessitated careful consideration of the mesh density and quality. Our detailed attention to meshing resulted in an improved representation of the flow physics around the vehicles, thereby a more accurate prediction of the aerodynamic forces. The results of this study affirm the pivotal role of vehicle design in influencing aerodynamics, fuel efficiency, and overall performance. It also demonstrates the power of CFD as a tool for design optimization in the automotive industry. Future research could expand upon this work by including more vehicle designs and exploring other aerodynamic parameters, offering further opportunities to enhance vehicle performance and efficiency through CFD analysis.

Contributions

	Name	Contribution
1	Muhammad Ahmed Shakeel	• CFD • Meshing
2	Muhammad Taha Khan	• CFD • Modelling • Conclusion/results
3	Sheikh Abdul Basit	• Literature Review

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